

LC 1340: Jump Game V

© Authored by Arpit Raj, LNMIIT Jaipur, August 13, 2026

Target Companies: Amazon, Facebook

Problem Understanding & Rules

In one step, we can jump from index i to $i + x$ or $i - x$.

We will be given a maximum jump length d such that:

$$0 < x \leq d$$

Jump Conditions

We can jump from index i to j only if:

- $arr[i] > arr[j]$
- And $arr[i] > arr[k]$ where $i < k < j$

Key Insight: So basically, index i must be the **tallest building** in a maximum range of d .

1. $|i - j| \leq d$
2. Every element between i and j is strictly smaller than $arr[i]$.

Goal: We need to return the maximum number of indices we can visit.

Visual Cases

The jumps can be visualized as jumping downwards to shorter buildings, bounded by taller buildings that block further jumps.

- **Case I:** A direct jump to a smaller building within range **d**.
- **Case II & III:** Jumping sequentially to strictly smaller buildings, maximizing the chain.

∴ We can reach the same location from multiple places, we need to use **Memoization**.

C++ Solution

```
class Solution {
public:
    int n;

    int dfs(vector<int>& arr, int d, int i, vector<int>& dp) {
        if(dp[i] != -1) {
            return dp[i];
        }

        int ans = 1; // khud ki building

        for(int j = i - 1; j >= max(0, i - d); j--) {
            if(arr[j] >= arr[i]) break; // poore range me i tallest hona chahiye
            // agar kahi bhi i se badi building mili to uske

            ans = max(ans, 1 + dfs(arr, d, j, dp));
        }

        for(int j = i + 1; j <= min(n - 1, i + d); j++) {
            if(arr[j] >= arr[i]) break; // poore range me i tallest hona chahiye
            // agar kahi bhi i se badi building mili to uske

            ans = max(ans, 1 + dfs(arr, d, j, dp));
        }

        dp[i] = ans;
        return dp[i];
    }
};
```

```
}

int maxJumps(vector<int>& arr, int d) {
    n = arr.size();
    int ans = 0;

    vector<int> dp(n, -1);

    for(int i = 0; i < n; i++) {
        ans = max(ans, dfs(arr, d, i, dp));
    }

    return ans;
}

};
```